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USER MANUAL

November 3, 2025

Geospatial Population Statistics Tool

```
Report of Selected World Places
==============================================================
121  northeastern places have a total population of  17650753
515  northwestern places have a total population of  57652493
3    southeastern places have a total population of     56007
36   southwestern places have a total population of   3677219
==============================================================
The Northwestern place of Fort Lauderdale has the highest population of 2042042
The Northwestern place of El Prevnar has the lowest population of 10
```

Overview

The Geospatial Population Statistics Tool is an add-on to QGIS that takes data on a selected set of points, categorizes them as being northeastern, southeastern, northwestern, or southwestern, then outputs the following information:

- Number of points allocated to each quadrant
- Total population of each quadrant
- The place(s) with the highest population
- The place(s) with the lowest population

This output is saved as a text file to a user-specified location.

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Parameters

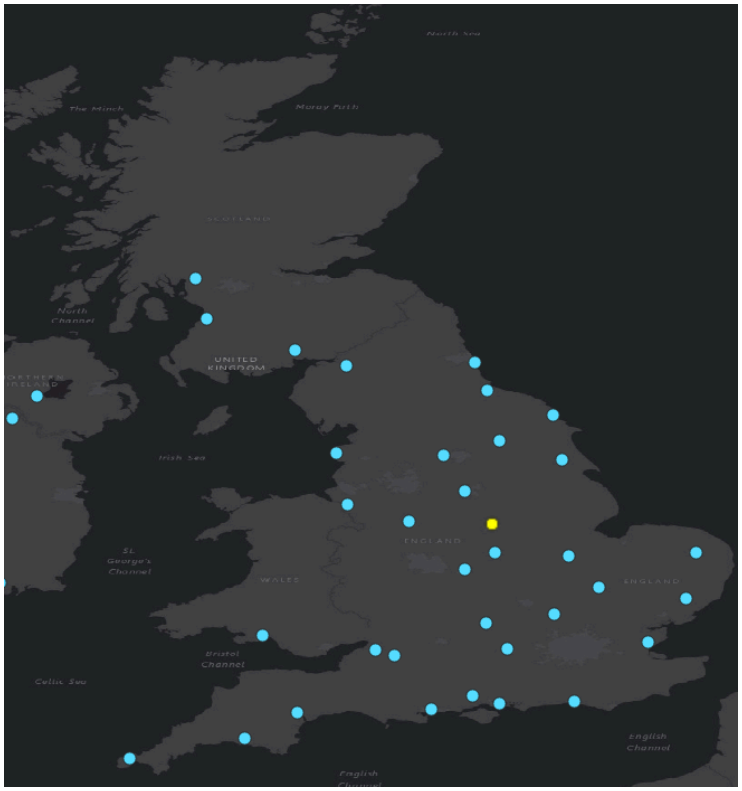
The Geospatial Population Statistics Tool requires 3 parameters:

- A dataset of points
- The selected points from the dataset
- The desired destination and name of the final text file

Details on each parameter can be found below:

Dataset of Points

The QGIS add-on requires a dataset to be loaded into your active QGIS pane. This dataset **MUST** be a collection of points with the name of the location as the first field of the attribute table, and the location's population as the second field. The points should be spatially located in a manner that is consistent with the real-world coordinates of the place in question. Consider the following example:



Here is a collection of points with one point (yellow) located in Nottingham being selected.

If we check the attribute table:

	nameascii	pop_max
1	Nottingham	825600

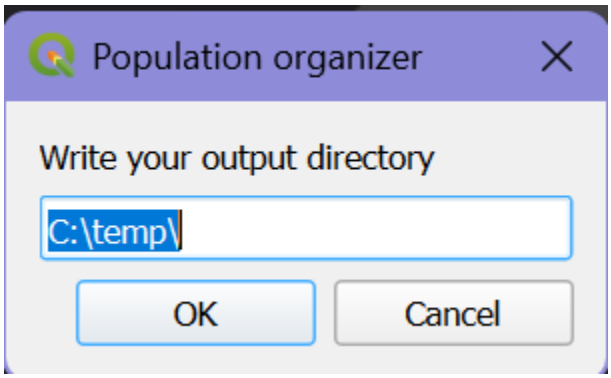
we see that the first column indicates the name of the place, and the second displays its population. The program requires a dataset with a minimum of two points, and does not have an upper limitation.

The Selected Points from the Dataset

The Geospatial Population Statistics Tool does not require all points within a dataset to be evaluated at once. Instead, the user may select whichever points they want to statistically analyze. Point selection must be done before running the add-on using QGIS-specific selection tools of the user's choice. If no points are selected, the add-on will analyze all points in the dataset.

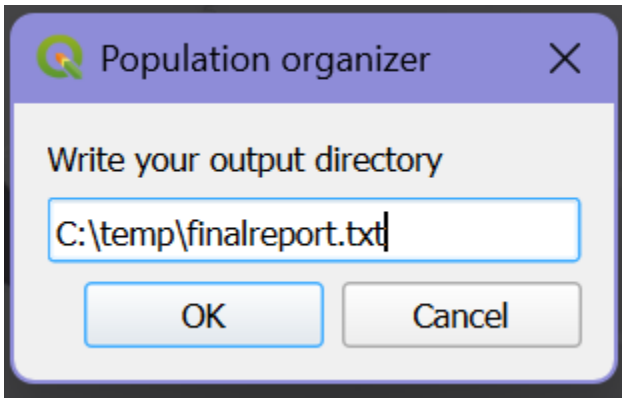
The Desired Destination and Name of the Final Text File

Once the add-on has finished processing the data, a popup will appear asking the user to specify where they would like the final report to be saved. The popup appears as follows:



The default directory links to the user's C:\temp\ folder, but may be altered as desired. If a folder is specified that does not yet exist, the folder will be created.

Within this popup, the user may also specify what name they would like to assign the final report file. Here is an example:



In this example, the filename will be ‘finalreport.txt’. When naming a file, specifying a file extension is not necessary: in this example, naming the file ‘finalreport’ or ‘finalreport.csv’ would lead to the file being automatically renamed to ‘finalreport.txt’. If the file already exists, ‘_NEW’ will be appended to the end of the filename.

Providing a filename is not mandatory: the add-on will default to naming the file ‘GlennGentilFinalReport.txt’. Should you want to provide a directory but not a filename, be sure to add a backslash (‘\’) or double backslash (‘\\’) to the end of the directory name. Failure to do so will name the file the final folder name. For example, inputting ‘C:temp\folder2’ would save a file called ‘folder2.txt’ to your temp folder, whereas writing the directory ‘C:\temp\folder2\’ would save ‘GlennGentilFinalReport.txt’ within ‘folder2’ of your temp folder. Leaving the input blank will default to writing the file to ‘C:\temp\GlennGentilFinalReport.txt’.

Once you are finished inputting the directory and filename, click ‘OK’ to save the report. Clicking ‘Cancel’ or the ‘X’ in the upper right corner will cancel out of the add-on and won’t save any files.

Determining Population Statistics

Each point is separated into one of four quadrants: Northeastern, northwestern, southeastern or southwestern, which will henceforth be allocated the abbreviations NE, NW, SE and SW for the rest of this user manual, respectively. This delineation is done by first identifying whether the point is north or south of the equator through analyzing whether its y-axis is above or below 0. Should a point rest squarely on the equator, it will be identified as being in the north.

Then a point is identified as being in the east or in the west by examining whether it is east or west or the prime meridian (whether its x-axis is above or below 0. Should a point rest squarely on the prime meridian, it will be evaluated as being eastern. Once these classifications are made, should a point be both northern and eastern it will be placed in the NE quadrant, northern and western, NW; southern and eastern, SE; southern and western, SW.

A running total is kept of how many points are within each quadrant, and the populations of all locations within any given quadrant are added up to give a total population for the quadrant.

Additionally, the points with the highest and lowest population counts are also tracked, and provided in the final report. If there is a tie for the location with the highest/lowest population, all locations are shown. If there is no tie, the designated quadrant of the location is shown.

Operating Instructions

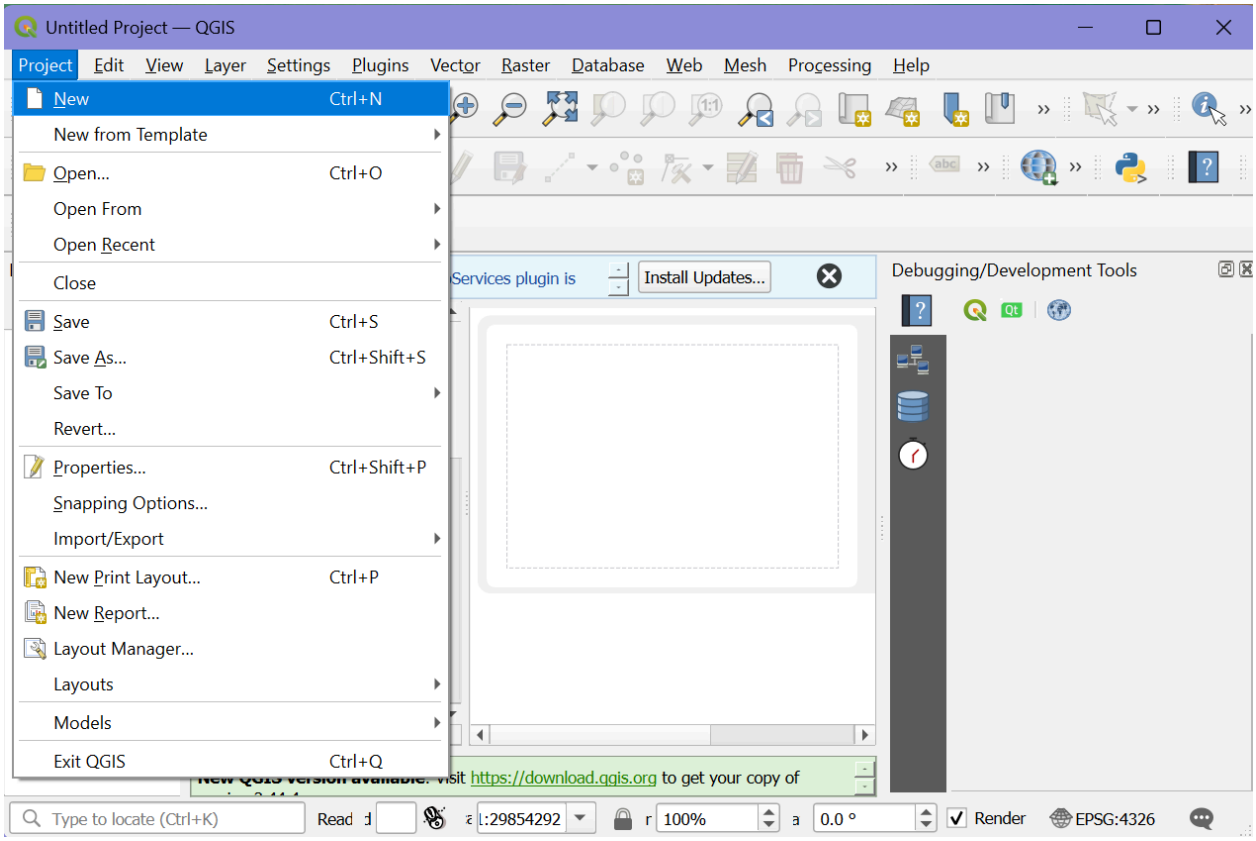
Here are step-by-step instructions for using the Geospatial Population Statistics Tool.

Step 1: Open QGIS

This add-on is designed to work within QGIS: a third-party piece of software. The Geospatial Population Statistics Tool and its affiliated creator do not claim any right or connection to QGIS or its developers. QGIS can be downloaded at the link below. This add-on was created on QGIS version 3.42.1 and may not work with other versions.

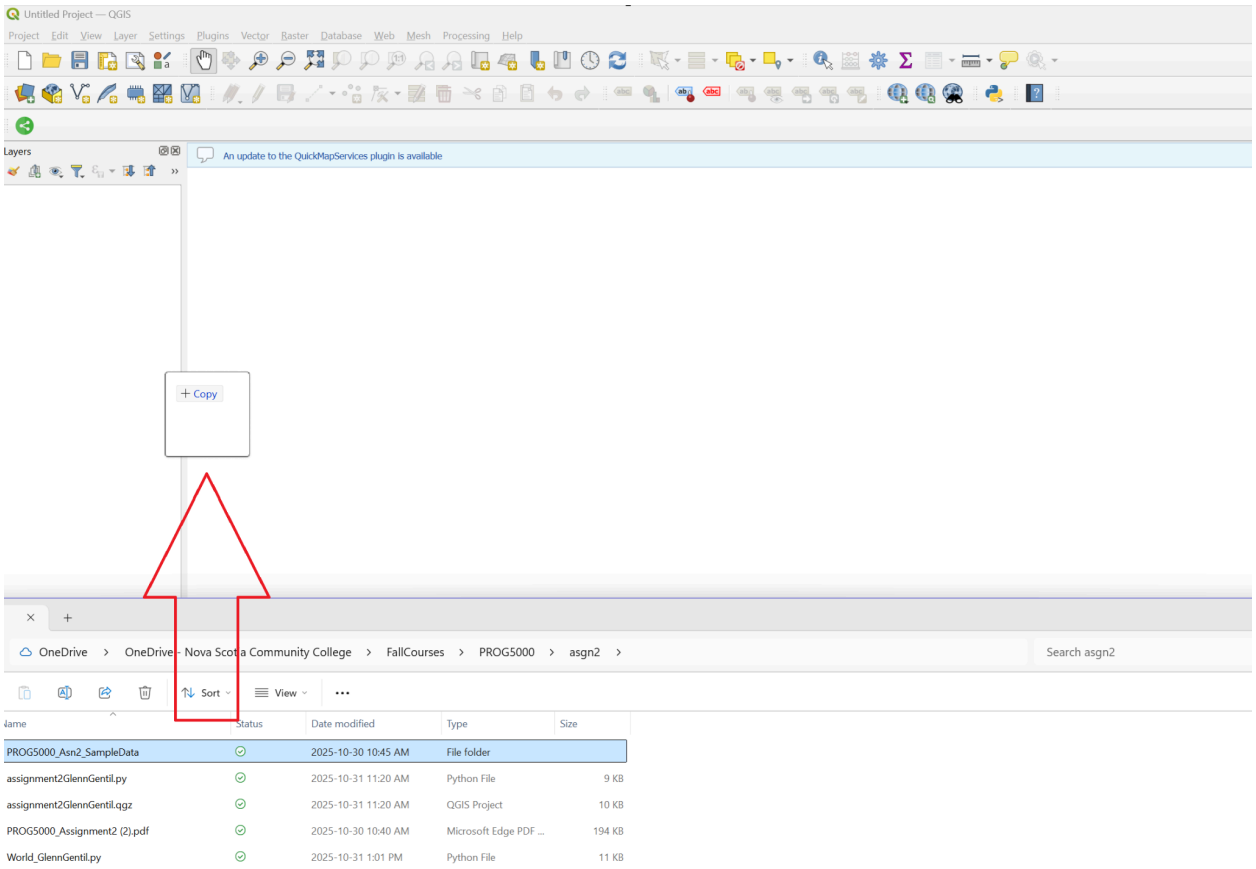
https://qgis.org/download/

Once downloaded, find where QGIS has been downloaded to your system and click to open it. Then click on ‘project’, then ‘new’.



Step 2: Adding your Data

Drag and drop your desired data into the new QGIS project you have created.



You should see your data in both the layers pane and the map pane.

Step 3: Integrating the Add-on

In your project, click the Python icon to open up the python console. From there, click show editor

then open script

A popup window will appear where you can select the location where you have the add-on downloaded. Once you have the add-on selected, click ‘open’.

Step 4: Using the Add-on

Select your dataset from the ‘layers’ pane, then select, from your imported dataset, however many points you would like to examine. If left blank, all points in the highlighted dataset will be used.

Click on the ‘run’ button in the python editor (**NOT** the python console), then input your desired file name and destination in the popup window. Details for this step can be found in the ‘parameters’ section of this user manual.

Once finished, your file will be found in the location you specified.

Usual Errors

The following are typical errors you may run into while using the add-on.

- object has no attribute 'selectedFeatureCount'

This error occurs when there is no dataset selected, or the dataset does not fit the required parameters. For information on what datasets are accepted, check the ‘parameters’ section of this user manual.

- FileNotFoundError: [WinError 3] The system cannot find the path specified:

This error occurs when a device, driver or network is specified that does not exist. Ensure the drive you are attempting to save your file to exists within your system.

Additionally, you may find that, when attempting to use the default filename, the file will be called ‘.txt’ as opposed to being called ‘GlennGentilFinalReport.txt’. This occurs when a forward slash (‘/’) is used to denote the final folder as opposed to a backslash (‘\’) or double backslash (‘\\’). Please be sure to use the backslash instead of the forward slash for denoting the file’s destination.

Limitations

The Geospatial Population Statistics Tool does well to provide general population statistics based on given points. That said, there are limitations to its capabilities, some of which are written below:

- The final report can only be saved as a .txt file
- The output layout/contents cannot be changed by the user in any way
- The place name and population are required to be the first and second fields, respectively
- The user cannot specify how NE, NW, SE and SW should be identified geographically

This list does not claim to be a complete set of all limitations of the add-on and other limitations may exist that are not included here.